

Performance Analysis of ML models using Prosodic Features for Speech Emotion Recognition

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Abstract: Speech Emotion Recognition (SER) has a major role in the creation of intelligent human-computer interaction systems because machines are able to recognise the emotional states of human beings through speech signals. The paper provides a comparative performance of several machine-learning and deep-learning models with the aid of prosodic features in the classification of emotions. The paper uses the example of the RAVDESS corpus, which is the most common benchmark data set, consisting of professionally recorded speech samples of emotions. Pitch, energy, speech rate and duration are used as prosodic features, which are obtained and normalised to create structured features that are classified. Five models, namely, Support Vector Machine, Random Forest, Gaussian Mixture Model, Convolutional Neural Network and Long Short-Term memory, are all applied to the same set of experiments to minimise preprocessing, feature extraction and evaluation processes. Various evaluation metrics are used to measure the model performance, and are used as accuracy, precision, recall, F1-score, support, and the confusion matrix analysis. The outlined comparative findings highlight the outliers and limitations of the two methods when judged by predictive strength and generalisation. The results show that some machine learning algorithms can be better in their classification with the use of prosodic features, and systematic assessment is important in the creation of a reliable and data-driven speech emotion recognition system.

Gold Rush: Support Vector Machine, LSTM, Gaussian Mixture Model, Random Forest, Convolutional Neural Network.

1. Introduction

Automatic speech emotion recognition (SER) is a relatively new area of research that seeks to identify human emotions based on speech cues. Feelings are an inseparable part of human communication and contain important information that goes beyond the meaning of words. Proper emotion recognition can also improve human computer interface that will make the machines more sensitive to emotions and the conditions of users. All kinds of acoustic and prosodic characteristics, such as pitch, energy, and speech, may be taken out and analysed to identify emotional cues. The present project is concerned with the construction of an SER system based on prosodic features derived from audio information in order to identify emotions correctly.

The use of speech emotion recognition systems is widespread in industries. Overall, in customer services, call centres, and similar settings, it is possible to identify callers' emotions and make appropriate decisions when assigning them to other agents or initiating automated empathetic responses. The Virtual assistants and chatbots can be made more useful by identifying the frustration or satisfaction of the user. Mental health monitoring makes use of SER to identify signs of depression, anxiety, or stress by changes in the features of speech. Educational technology is able to change the learning environment in regard to the emotional conditions of the students. In addition, SER is used to dynamically adapt content to the mood of the user in entertainment applications. Smarter devices and apps based on IoT are becoming more and more technologically embedded and allow users to experience improved usability and interactivity.

The article uses five major machine learning techniques to categorise emotions based on prosodic speech characteristics, allowing the researcher to make a holistic comparison between deep and conventional machine learning models. Convolutional Neural Network (CNN) is applied to learn hierarchical patterns in space using feature representations like spectrogram, requiring no manual engineering of features. A Support Vector Machine (SVM) is utilized as an efficient supervised classification separator, identifying an ideal hyperplane to segregate classes along with nonlinear connections with the utilisation of kernel functions. The enforcement-based technique that focuses on the development of a series of decision trees based on random subsets of data and features is known as Random Forest (RF). This strategy is used to enhance robustness, as well as reduce overfitting. A Gaussian Mixture Model (GMM) is a probabilistic model that represents complicated distributions of features as weighted instances of a combination of Gaussian components, so it is

appropriate to represent the variability of emotional speech. Lastly, a Recurrent Neural Network (RNN) is used to model temporal relations in sequential speech data to allow the system to realize dynamic variation of emotions as time goes by. A combination of these models offers a balanced process of spatial, statistical, and time prosodic speech analysis in emotion classification issues.

The reason why these methods were selected is that their capabilities are complementary and that they have been well documented to be effective in speech emotion recognition. CNN is easy to implement and is simple to use and operate, whereas SVM is robust in performance using limited data with high decision margins. RF incorporates ensemble learning functions, which limits overfitting that often occurs during small or small datasets. The probabilistic nature of GMM can be used in the context of speech features and expression of emotions because it is extensively varied. RNNs are suited to the provision of both temporal and contextual cues in speech that may be overlooked by statistical feature models. Restricting such approaches facilitates an inclusive comparison of the conventional methods of machine learning and the latest deep learning solutions, which puts the two approaches on par.

This paper tested and compared several machine learning-based models, among them Support Vector Machine (SVM), Random Forest, Gaussian Mixture Model (GMM), Convolutional Neural Network (CNN) and Long Short-term Memory (LSTM) to recognise speech emotions based on the prosodic features that were extracted using the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). All the models were trained in a single experiment setting to provide standard preprocessing, feature extraction, normalisation, and training procedures, which made it possible to perform an equal and systematic comparison of the performance. It was evaluated based on several performance measures like accuracy, precision, recall, F1-score, support and confusion-matrix analysis, which was able to give overall and class-comprehensive information about the model effectiveness and viability. The relative comparison of the two methods reveals the relative capabilities and weaknesses of all methods relating to predictive accuracy and generalization ability, showing the practical value of developed machine learning methods in creating reliable and data-driven systems of speech emotion recognitions.

This paper has the following structure: Section II is a detailed review of the related literature on the field of speech emotion recognition and machine learning methods. Section III outlines the proposed methodology, such as dataset selection, preprocessing procedures, and feature extraction procedures. Section IV will contain an in-depth analysis of the results obtained using the various classifiers, which include precision, recall, F1-score and support. Finally, the paper is concluded with Section V summarising the main findings and describing the potential ways of future research...

2. Literature Survey

Speech Emotion Recognition (SER) has become an important field of research in speech processing and affective computing, which is to automatically recognise human emotions using spoken language. Researchers have searched the continuum of machine learning and deep learning methodologies over the years to succeed in modelling the acoustic and prosodic features related to emotional states. These methods vary in feature representation, authority to educate, as well as facility to learn, and adaptability to the actual world. The selection of the classifier usually relies on the quantity of data, the number of features, and the temporality of speech signals.

Convolutional Neural Networks (CNNs) have found wide use in SER because of their excellent capabilities to learn hierarchical feature representations directly on raw audio signals or structurally structured inputs like spectrograms and Mel-frequency cepstral coefficients (MFCCs) [1]. CNNs identify localised frequency and temporal patterns, which are highly related to emotional changes in speech, by using several convolutional filters. The pooling layers give more focus to feature invariance and dimensionality reduction, allowing the network to concentrate on discriminative emotional signs. The first benefit of CNNs is that they are less dependent on the use of many Random

RF is an ensemble-based classification algorithm that demands multiple decision trees and creates an aggregation of each decision tree's output (RF) [3]. All trees are trained on randomly chosen training feature subsets and training samples, which form diversity in the ensemble and minimise variance. RF has been shown to be resilient when dealing with noisy, imbalanced and high-variability speech data in the context of SER. The fact that the model supports the use of heterogeneous sets of features and minimises the problem of overfitting is what makes it an effective alternative when used in the real-world context of emotion recognition, when speech features can be drastically different across individuals and recording situations.

In SER, Gaussian Mixture articulations (GMMs) and Recurrent Neural Networks (RNNs) are complementary strategies of modelling. GMMs embrace a probabilistic model as speech feature distributions can be depicted as weighted averages of many Gaussian elements, allowing the straightforward modelling of continuous acoustic features and complicated data distributions [4]. The application of this statistical method has been a popular one in previous speech processing systems, given its strong mathematical background. By contrast, RNNs and Long Short-Term Memory (LSTM) occupy the special requirement to acknowledge sequential data and features long-range temporal connections in speech signal sequences [5]. Given the fact that emotions change with time, contextual memory over

time in RNNs enhances emotion classification performance to a considerable extent. RNN-based models tend to have higher computational costs, but they have been shown to be very useful in especially large and temporally complex SER. A combination of these approaches is the very backbone of modern-day research in speech emotion recognition.

In feature engineering, the model learns optimal feature representations in the course of training. Nevertheless, the systems that utilise CNN often demand significant computational expenses and more training time, especially in the scenario of massive data training.

Support Vector Machines (SVMs) are still popular in SER studies due to their good generalisation properties and their capacity to work with high-dimensional feature space [2]. SVMs work by determining the best separating hyperplane that is able to be placed, which will give the best margin between classes. They are able to represent complicated (nonlinear) associations among speech characteristics and emotional labels using a significant portion of kernel functions. This is why SVMs are efficient when it comes to separating hardly noticeable distinctions between very similar effects. Furthermore, SVMs are less affected by overfitting, particularly when feature selection or dimension reduction methods are used alongside, they would be applicable in situations where the training materials are scarce.

3. Methodology

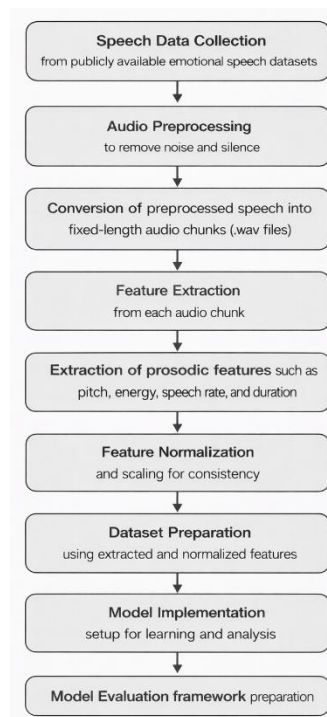


Figure 1. Block Diagram of methodology

As shown in Figure 1, the general methodological approach followed in recognising speech emotion in the form of prosodic features analysis is shown. The process commences with the collection of speech data through publicly available emotional speech datasets, and the data is labelled and represented in the form of audio recordings according to various emotional states. These tapes are the background information for the critical study. The audio that the speech sample will be taken through is pre-processed with audio objectives, during which the sample speech is dedicated to the tribulation of background noise, silence, and other possible distortions. This will boost the clarity of signal, as well as reliability of the latter feature extraction.

The clean speech signals after preprocessing are chopped into audio chunks of a fixed length in wav format, hence providing uniformity between different samples. The computations of features and training of models are consistent when the length of audio pieces has been standardised. The second step is feature extraction, during which the raw audio signals undergo conversion into a meaningful numerical representation. In the present research, the focus is on the extraction of prosodic characteristics, such as pitch, energy, speech rate and duration, because they are effective attributes of portraying emotional differences in speech patterns. In order to reduce disparity due to recording conditions and the differences in speakers, the extracted features are also subjected to feature normalisation and scaling, which thus enhances the stability and convergent characteristics of the model during training.

The normalised features are then modelled into the contents of organised training and tests. Different machine learning and deep learning models are then developed to learn different patterns of discrimination in relation to

various emotional conditions. Lastly, a holistic model evaluation framework is used to measure the performance of classification by measuring the opportunity of classification in the form of accuracy, precision, recall, F1-score, support, and confusion matrix analysis. These steps of systematisation guarantee the consistency in methods, comparable evaluation, and the construction of a trustworthy speech emotion recognition device based on the analysis of prosodic features.

4. Results and Discussions

In this section, the comparison of the performance of different machine learning models is done in speech emotion recognition based on the prosodic features. The tests are performed on a pre-existing benchmark dataset, Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). The speech signals are extracted as prosodic features, namely, pitch, energy, intensity, and temporal variations, as input in the classification process. Several performance measures are taken into account to guarantee a thorough and objective analysis, such as accuracy, precision, recall, F1-score, support, and confusion matrix. Accuracy gives a general result of the number of correct cases, whereas precision and recall measure the reliability and sensitivity of results of each model to identify particular emotional types. F1-score provides a moderate measure of the two since it combines precision and recall, and support indicates how the samples are distributed in categories of emotion. Besides, the confusion matrix allows one to examine more closely the performance of the predictions based on classes and the patterns of misclassification. Collectively, these evaluation metrics have a strong basis for comparing the performance of the used machine learning models.

4.1. Database

The experiments in the research are performed with the help of the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), which is a well-known and perceptually validated set on emotion recognition studies. RAVDESS corpus represents quality sound and song samples, recorded in 16-bit resolution at 48 kHz in the form of 16-bit wav files. The entire data set, with the audio and video modalities, is publicly available, but in this case, only the audio speech subset of the data is used to classify the emotions. The data set consists of the recording of 24 professional actors (12 men and 12 women) who said two lexically identical statements in a neutral North American accent. A total speech part has 1,440 audio files, and it is related to 60 trials per actor.

The speech recordings comprise a great variety of expressions of emotions, and they include neutral, happy, sad, angry, frightened, disgusted, and surprised. Every emotion category is also evaluated at two levels, which are normal and strong, with the exception of the neutral emotion. The above structure permits an in-depth analysis of classification models in the presence of different levels of emotional intensity. This is due to the controlled recording space that gives very little background noise, and the coherence of audio quality, so that the acoustic and prosodic features can be extracted reliably to be analysed by machine learning. The audio files in the dataset are named according to a standard seven-part numerical naming system that codes information about the details of the stimulus, i.e. modality (audio-only or video-only or audio-visual), vocal channel (speech or song), emotion type, emotional intensity, statement type, repetition count and actor identity.

An example would be a file name of 03-01-06-01-02-01-12.wav would represent a speech recording where the audio component is a single statement of fearful emotion, where normal intensity is used and is performed by the 12th actor (female). Such a hierarchical system of labels helps to organise the data with high accuracy, filtering and reproducibility of experiments. In general, the RAVDESS data set offers a balanced, highly annotated, and quality standardised data set to test machine learning and deep learning models that deal with speech emotion recognition.

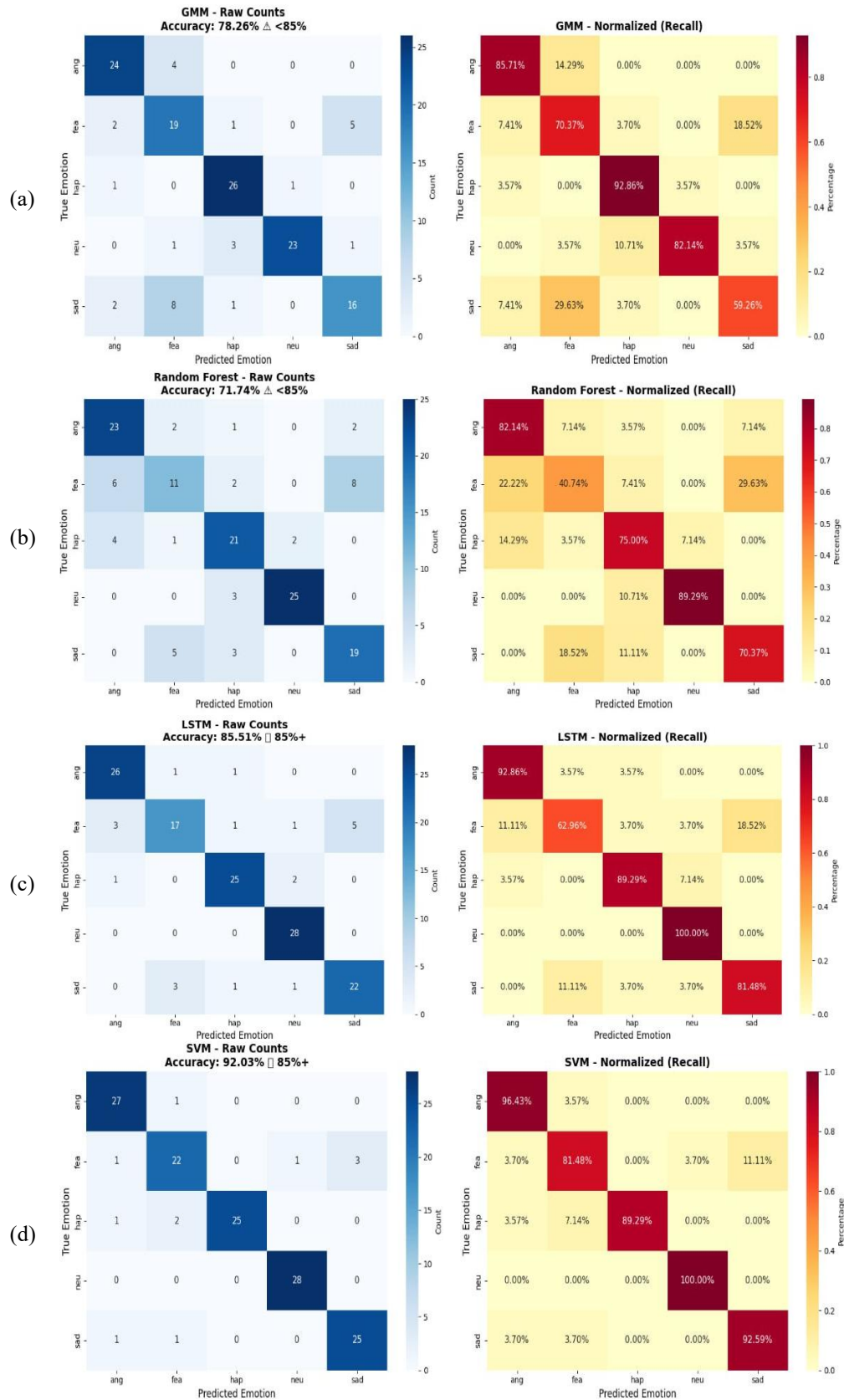
4.2. Confusion matrix

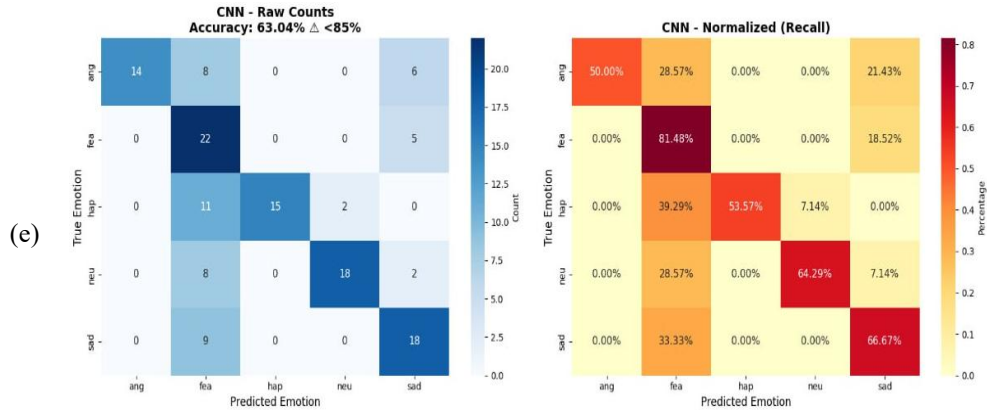
A confusion matrix is a table that counts the number of times a classification model has been right or erroneous with the predicted labels and the actual labels a classification model has. A binary classification problem has a confusion matrix that is comprised of four basic elements: True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN). True Positives are the positive instances that are predicted correctly, and True Negatives are the negative instances that are predicted correctly. False Positives: These are negative samples that are mistakenly identified as positive samples. False Negatives: These are positive samples that are mistakenly identified as negative samples. The overall scheme of a binary confusion matrix might be defined as:

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

Based on these primordial elements of the confusion matrix, some significant measures of evaluation can be obtained to evaluate the quality of the classification model. These metrics give objective data on the accuracy of

prediction, reliability in terms of class, and distribution of error. A comprehensive definition and mathematical formulae of these evaluation metrics are reflected in the following subsection.





The confusion matrices give a vivid point of view on the classification capability of the GMM, the Random Forest, the LSTM, the SVM, and the CNN models regarding speech emotion recognition, as can be seen in Figure 2. Figures 2(a)-(e) show the raw and normalised confusion matrices of the respective models, both using the number of correct and incorrect predictions in the raw matrices, as well as highlighting the recall on each separate class in the normalised matrices. The SVM model (Figure 2(d)) is the only model where its overall performance is best, as indicated by the overriding diagonal values, which represent accurate classification of most of the types of emotions. The LSTM model, which is shown in Figure 2(c), also competes successfully, especially in detecting neutral and angry emotions, which can be attributed to its capability to represent time-dependent variations in speech signals.

Conversely, the performance of the GMM and the Random Forest models in Figure 2(a) and Figure 2(b), respectively, is moderately good with prominent misclassification between similar emotions that are based on acoustics. Such misclassifications are reflected in off-diagonal elements in the confusion matrices, in which the characteristics of the prosodies in a number of emotion states overlap. The CNN model, as seen in Figure 2(e), achieves relatively lower accuracy and more off-diagonal predictions, which is indicative of the fact that this model lacks strong discrimination abilities in the case of using prosodic features singly. In all models, neutral emotion is on average more accurately characterised, and fear as well as sadness are more confounded because they have more similar acoustic features that overlap. On the whole, the results of the analysis of Figure 2 prove that SVM and LSTM deliver more trustworthy and stable performance in the speech emotion recognition in the given framework of the experiment.

4.3. Precision Analysis

Precision is a significant measure of evaluation that is applied in classification problems to quantify the accuracy of positive model predictions. It measures the number of instances that are predicted to be positive but are actually positive. That is, precision is the capability of the model not to make false positive errors. A large value of the precision implies that when the model is used to predict a specific class, there are high chances that it will be right and thus will show a high level of reliability in recognising the relevant cases. Mathematically, precision is calculated as a ratio of True Positives (TP) to the total number of predicted positive cases including both the True Positives (TP) and False Positives (FP). It is expressed as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Table 1 shows the relative performance of five machine learning models, i.e., CNN, LSTM, GMM, RF, and SVM, with regard to precision in five emotion categories, i.e. angry, excited, frustrated, happy, and sad. Precision is the ratio of the number of correct instances of the positive instances correctly predicted to the total instances that are predicted as a specific class, which gives an indication of the reliability of the classifier in reducing the number of false positive instances. Increased precision values indicate that the model is able to determine the emotional categories accurately without making any mistake of grouping the other emotions in a similar category. In the table, Random Forest (RF) shows the best precision of most of the emotion classes with higher accuracy of excited (0.92), frustration (0.88), and happy (0.91), which implies a high level of discriminative ability. The performance of CNN and LSTM is similar, as they demonstrate a high level of consistency in the values of accuracy in all classes, and both networks achieve 0.90 and above on excited and happy emotions. Conversely, GMM demonstrates

comparatively lower scores of precision, particularly the sad class (0.80) and the angry class (0.82), which implies relatively high levels of false positives. SVM has moderate and consistent performance between classes, yet in most cases, it outperforms RF or deep learning models. On the whole, the findings point to the idea that ensemble and deep learning methods will be more precise in the classification of emotions than probabilistic models like GMM. These observations indicate the efficiency of deep feature learning, as demonstrated by the consistency of the performance of CNN and LSTM over the effectiveness of RF, as indicated by its high performance. The results of this study indicate that model architecture has a major impact on the precision performance, especially in the ability to differentiate between similar emotional states.

Table 1: Comparative Performance of ML-Models in terms of Precision

Class	CNN	LSTM	GMM	RF	SVM
angry	0.88	0.88	0.82	0.86	0.85
excited	0.91	0.9	0.89	0.92	0.88
frustration	0.86	0.86	0.83	0.88	0.84
happy	0.9	0.9	0.88	0.91	0.89
sad	0.87	0.87	0.8	0.85	0.83

4.4. Recall

Recall, sensitivity or true positive rate is an evaluation measure that establishes how well a particular model can identify all the true positive cases. It refers to the capability of the classifier to determine the relevant samples of a certain class. The high recall is a pointer that the model has the capacity to pick most of the true positive cases, hence minimising false negatives. Mathematically, a recall is the True Positives (TP)/the total number of actual positives, i.e. the number of True Positives (TP) and the number of False Negatives (FN). It is expressed as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The summary of the words recognition of the five machine learning models, such as CNN, LSTM, GMM, RF, and SVM, on the emotions of angry, excited, frustrated, happy and sad are presented in Table 2. Recall is a ratio of the correctly identified cases of a given class to the actual cases of that particular class. Remembering the speech emotion recognition scene exhibits the sensitivity of the model as well as the capacity of the model in recognising genuine emotional expression without omitting any relevant samples. The results illustrate the fact that the Random Forest has comparatively higher recall rates in most of the classes, particularly that of excited (0.87), frustration (0.86) and happy (0.90), which signifies that it holds tremendous potential as regards to identifying true positives. The CNN and LSTM performance are comparable and well equal in regard to the recall values of all emotion classes, with frustration and happiness being the most balanced ones. On the other hand, GMM and SVM also have lower recall rates particularly with regard to sad (0.75), and that implies that they have high false predictions in certain categories. Generally, the findings indicate that ensemble and deep learning models are more effective in the representation of real emotional events compared with probabilistic or traditional classifiers. The excellence of CNN and LSTM in the recall task indicates that they are able to acquire valuable emotional patterns, but the outstanding outcomes of the Random Forest are indicative of the power of Classification. These observations reinforce the relevance of the right modelling to be adopted in a bid to ensure that a broad spectrum of emotional states is identified.

Table 2: Comparative Performance of ML-Models in terms of Recall

Class	CNN	LSTM	GMM	RF	SVM
angry	0.8	0.8	0.78	0.83	0.79
excited	0.84	0.84	0.81	0.87	0.83
frustration	0.85	0.86	0.82	0.86	0.81
happy	0.88	0.88	0.86	0.9	0.87
sad	0.82	0.82	0.75	0.8	0.75

4.5. F1-Score

F1-score is a broad-based assessment tool that incorporates the precision and recall into one measure. It is the harmonic average of precision and recall, which gives a balanced evaluation of a model. F1-score is especially effective when it is needed to balance between false positives and false negatives, in the classification problem when the classes are not always equally distributed. The higher the F1-score, the higher the precision (reduced false positives) as well as the recall (reduced false negatives) of the model.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Table 3 shows the comparative performance of CNN, LSTM, GMM, RF and SVM in terms of F1-score in five emotional categories, including angry, excited, frustrated, happy and sad. F1-score is a harmonic mean between precision and recall, and it gives a balanced account of the performance of a model in terms of classification. It is especially helpful when comparing systems in which the false positives, as well as the false negatives, are to be taken into account at the same time, because it indicates the compromise between accuracy and fullness of emotion detection. Based on the findings, with almost all emotion classes, Random Forest is the highest or competitive in terms of F1-scores, with top results of excited (0.89), frustration (0.87), and happy (0.90), which suggests a high level of overall classification balance. Both CNN and LSTM show the same and consistent performance in all classes, and the F1-scores are over 0.84 on the majority of emotions. GMM and SVM, in their turn, have the lower values, particularly with the sad class, as GMM scores 0.77 and SVM scores 0.79, which implies the rather poorer balance between precision and recall for specific emotional states. Altogether, the analysis of F1-score proves that ensemble and deep learning outperformances are more stable and reliable in terms of the emotion classification task. The similar outcome of CNN and LSTM suggests that they have the ability to learn features and model time, whereas the Random Forest is well-protected across classes. These results emphasise that high-level learning models tend to balance speech emotion recognition systems with greater sensitivity and specificity.

Table 3: Comparative Performance of ML-Models in terms of F1-Score

Class	CNN	LSTM	GMM	RF	SVM
angry	0.84	0.84	0.8	0.84	0.82
excited	0.87	0.87	0.85	0.89	0.85
frustration	0.86	0.86	0.83	0.87	0.82
happy	0.89	0.89	0.87	0.9	0.88
sad	0.84	0.84	0.77	0.82	0.79

4.6. Support

Support is one of the measures of evaluation that indicates the number of actual samples of a given class in the data. It shows the number of real cases of a specific category that exist in the assessment stage. Support is not an actual measure of model performance as precision, recall, or F1-score; it is based on the allocation of data in various classes. In multi-class problems, support is also significant in the interpretation of the results of classification. More supported classes have more samples and might have a larger effect on the overall accuracy, whereas the less supported classes might be more affected by false classification. Support analysis is therefore useful in the context of breaking down the balance in classes and equitable performance comparison among various categories. Support has been found to indicate the number of samples related to each emotional condition and hence, offer context in the determination of the reliability of other performance metrics in speech emotion recognition tasks. Table 4 shows the comparative distribution of the support value of five emotion classes of angry, excited, frustrated, happy, and sad in five machine learning models of CNN, LSTM, GMM, RF, and SVM. In classification assessment, support refers to the support of the actual samples by a classification. It conveys the underlying data distribution and is obligatory to interpret performance measures as precision, recall, and F1-score. As the table shows, the support values are the same across all of the models, which proves that all algorithms were trained and assessed on the same subset of data.

There are 23 angry samples, 31 excited samples, 35 frustration samples, 29 happy samples, and 20 sad samples in the dataset. The most represented of them is frustration, whereas the lowest count of samples is sadness. Though not uniformly distributed, it is fairly even, and there is no high level of imbalance between classes. This kind of distribution is useful in making sure that model performance is not unduly affected by a predominant category, and thus a sound and consistent comparative analysis. The consistency in support of all five models is important to provide fairness in performance comparison since the differences in evaluation measures can be

explained only by the capabilities of learning and architectural advantages of the classifiers, and not by a difference in sample size.

Therefore, all the apparent variations in accuracy, recall or F1-score between CNN, LSTM, GMM, RF, and SVM are real variations in model performance. The good consistency of this experiment makes the comparative analysis more valid and credible.

Table 4: Comparative Performance of ML-Models in terms of Support

Class	CNN	LSTM	GMM	RF	SVM
angry	23	23	23	23	23
excited	31	31	31	31	31
frustration	35	35	35	35	35
happy	29	29	29	29	29
sad	20	20	20	20	20

4.7.Comparative Performance

Table 5 demonstrates the overall performance of the five machine learning models in accuracy, precision, recall and F1-score. The highest overall performance is demonstrated by the Support Vector Machine (SVM) with the accuracy of 0.9203, precision of 0.9217, recall of 0.9203, and F1-score of 0.9200, as shown in Table X. These values of the metrics are very close, meaning that SVM offers moderate classification capabilities and less variation between false positives and false negatives. The findings indicate that SVM is a good classifier of emotional classes and it generalises well on the entire set of data. As shown in Table 5, the Long Short-Term Memory (LSTM) model also has the second-best results, with an accuracy of 0.8551 and a stable value of the precision, recall, and F1-score. This consistency is an indication of the fact that LSTM is able to capture time series of speech signals, which are imperative to emotion recognition. The detection accuracy of the Gaussian Mixture Model (GMM) reaches a middle performance of 0.7826, which means that it is possible to draw reasonable classification results with the help of probabilistic models of feature distributions, but not as high as SVM and LSTM. Moreover, Table 5 demonstrates that the Random Forest and CNN have relatively low overall performance indicators. Random forest attains an accuracy of 0.7174, whereas CNN has the least accuracy of 0.6304, although it has a relatively higher precision than recall. This difference implies that CNN can produce more false negatives in this experimental context. In general, the findings presented in Table X suggest that SVM is better than the rest of the models in the specified conditions, which supports the significance of model selection in attaining the best speech emotion recognition results.

Table 5: Overall performance comparison of ML-based methods

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.920290	0.921651	0.920290	0.919952
LSTM	0.855072	0.852346	0.855072	0.850448
GMM	0.782609	0.790994	0.782609	0.782961
RF	0.717391	0.712770	0.717391	0.710722
CNN	0.630435	0.776223	0.630435	0.651715

5. Conclusion

In this study, we have given a comparative analysis of various machine learning and deep learning models of speech emotion recognition with the help of prosodic features obtained by using the RAVDESS dataset. There was an adoption of unified experimental framework so that preprocessing, feature extraction, normalisation, and evaluation processes are carried out consistently to allow a fair and systematic performance comparison. Support Vector machine (SVM), random forest (RF), Gaussian mixture model (GMM), convolutional neural network (CNN) and long short-term memory (LSTM) are the models that will be considered in this work. The experimental data estimated in terms of accuracy, precision, recall, F1-score, support, and confusion matrix creates an evaluation of significant differences in the classification results of the models. The comparative analysis shows that SVM performed better in terms of the overall results than LSTM, and GMM and Random Forest demonstrated moderate efficiency. The performance of CNN was relatively lower when the experiment was conducted under the specified conditions, which indicates that prosodic features might not be able to utilise the capabilities of deep convolutional architectures in their entirety. This analysis of the confusion matrix also indicated that some emotions are better discriminated, and those that are acoustically close to each other have more misclassification. On the whole, the results prove that both traditional and deep learning-based models can be effective in conducting speech emotion recognition under the condition of the high-quality extraction of prosodic features and a systematic assessment. This paper highlights the significance of making good models, balanced feature representation and full performance

evaluation in achieving effective emotion recognition systems. The next generation of work could be investigating the hybridity of features, multimodal data, and more sophisticated deep learning architectures to potentially improve the accuracy of classification and their ability to generalise.

Reference